PyGameMaker — Tutorial 4: Breakout

How to Use This Handout

This handout goes with the in-app tutorial **Help > Tutorials > Breakout: Brick Breaker** (7 pages, about 25-30 minutes). Keep the tutorial open on one half of your screen. Tick each box when the step is done, and check the **You should see** line at the end of every phase. If something does not match, look at **Stuck?** before you call your teacher.

What You Will Make

The classic brick breaker: a paddle at the bottom, a ball that bounces off walls and the paddle, four rows of coloured bricks to destroy, three lives and a score. You will use a **parent object** so one collision rule works for every brick colour.

Phase 1: A Bouncing Ball

- [ ] **1.** Create four sprites: spr_ball (16×16), spr_paddle (64×16), spr_wall (32×32) and spr_death_zone (32×32). Draw them or import the sample images from Tutorials/04_breakout/assets/.
- [ ] **2.** Create the objects: obj_wall_side and obj_wall_top (sprite spr_wall, **Solid**), obj_paddle (**Solid**), obj_ball (**not** solid), and obj_death_zone (sprite spr_death_zone, **Visible unchecked**).
- [ ] **3.** Paddle: **Keyboard: Left Arrow (held)** with *Set horizontal speed to -8*, **Keyboard: Right Arrow (held)** with *Set horizontal speed to 8*, **Keyboard: No key** with *Stop movement*, and **Collision with obj_wall_side** with *Stop movement*.
- [ ] **4.** Ball: **Create** with *Set horizontal speed to 3* and *Set vertical speed to -3*.
- [ ] **5.** Ball collisions: with obj_wall_side → *Reverse horizontal movement*; with obj_wall_top → *Reverse vertical movement*; with obj_paddle → *Reverse vertical movement*; with obj_death_zone → *Jump to x: 320 y: 100*.
- [ ] **6.** Create room_breakout (640×480). Place side walls down both edges, top walls along the top, the death zone along the **bottom** (there is no bottom wall), the paddle near the bottom centre and the ball just above it.

> **You should see:** press **F5**. The paddle moves with the arrow keys and stops at the walls. The ball bounces around; if it falls past the paddle it comes back at the top.

Phase 2: First Row of Bricks

- [ ] **7.** Create a sprite spr_brick_red (32×16).
- [ ] **8.** Create obj_brick_parent (no sprite, **Solid**) — a template. Then create obj_brick_red (sprite spr_brick_red, **Parent** obj_brick_parent, **Solid**).
- [ ] **9.** In obj_ball, add **Collision with obj_brick_parent**: *Reverse vertical movement*, *Destroy other instance*, *Add to score 10*.
- [ ] **10.** In the room, fill one row of obj_brick_red near the top (turn on **Snap to Grid** 32×16).

> **You should see:** press **F5**. The ball destroys bricks and bounces back. Each brick is worth 10 points (you cannot see the score yet).

Phase 3: More Brick Rows

- [ ] **11.** Create sprites spr_brick_orange, spr_brick_yellow and spr_brick_green (32×16).
- [ ] **12.** Create obj_brick_orange, obj_brick_yellow and obj_brick_green, each with its sprite, **Parent** obj_brick_parent and **Solid**. Do **not** change the ball.
- [ ] **13.** Fill four rows: red, orange, yellow, green.

> **You should see:** press **F5**. Bricks of every colour break when the ball hits them.

Phase 4: Game Controller

- [ ] **14.** Create obj_game_controller (no sprite). **Create**: *Set lives to 3*, *Set score to 0*. **Draw**: *Draw score at x 10, y 10* and *Draw lives at x 200, y 10*.
- [ ] **15.** In obj_ball, in the **Collision with obj_death_zone** event, add *Add to lives -1* **before** the jump.
- [ ] **16.** In obj_game_controller, add the event **Other Events > No More Lives** with *Show message "Game Over!"*, then *Show Highscore*, then *End Game*.
- [ ] **17.** Place obj_game_controller anywhere in the room.

> **You should see:** the score and lives show at the top. Each fall costs a life. After the third fall you get a Game Over message, the highscore table, and the game closes.

Stuck?

What is wrong	Most likely cause	What to do
The ball flies out of the room	A wall is missing or not Solid, or the ball has no bounce event for it	Check the walls; check the ball's collision events
The ball bounces but the bricks stay	The brick collision event is on the wrong object, or bricks do not have the parent set	Put the event on obj_ball for obj_brick_parent; set the **Parent** on every brick
Only red bricks break	The new bricks have no **Parent**	Set **Parent** to obj_brick_parent on orange, yellow and green
The paddle goes through the side walls	No **Collision with obj_wall_side** event on the paddle	Add it with *Stop movement*
The ball is never lost / lives do not go down	The death zone is missing, or *Add to lives -1* was not added	Place the death zone at the bottom; add the action
No score or lives on screen	obj_game_controller is not in the room	Place it in the room
Lives start at 0 or the game ends at once	*Set lives to 3* is missing	Add it to the controller's **Create** event
The ball goes through a brick without breaking it	The brick is not Solid or not in the room properly	Tick **Solid** on the brick objects

Challenges

- **Try this (5 minutes):** change the ball speed or the paddle speed.
- **Push further:** give each colour a different score (red 40, orange 30, yellow 20, green 10), or speed the ball up as bricks disappear.
- **Invent:** build a second level with a different brick layout, or play a sound when the game ends.

Vocabulary

Term	What it means
Parent object	A template object; objects that use it inherit its events
Child object	An object that has a parent (each brick colour)
Death zone	An invisible object that detects when the ball is lost
Lives	How many times you can lose the ball
Highscore	The best scores, kept in a table

Check Yourself

Write your answers in the notes below.

1. Why is the collision event written for obj_brick_parent and not for each colour?
2. Why is there no wall at the bottom of the room?
3. What does *Reverse vertical movement* do, and why is it used for the top wall and the paddle but *Reverse horizontal* for the side walls?

My Notes